

Project Report: Route4Me - Delivery Route Optimization System

1- Introduction

Route4Me is a C++ program designed to optimize package delivery routes by computing:

- **Minimum Spanning Tree (MST)** using **Prim's Algorithm** (for cost-efficient network design)
- **Shortest Paths** using **Dijkstra's Algorithm** (for fastest delivery routes from any zone)

2- Features

Interactive Menu System – Allows multiple operations in one run

Dynamic Route Management – Add/remove routes and recalculate optimizations

Graph Visualization – Displays current routes in adjacency matrix format

Error Handling – Validates all user inputs

3- Algorithms Used

Prim's Algorithm

- Finds the **minimum spanning tree** (MST) to connect all zones at the lowest cost.
- Output edges and total minimum cost.

Dijkstra's Algorithm

- Computes **the shortest paths** from a given source zone to all others.
- Displays distances and exact paths.

4. Implementation Details

- **Data Structures:**
 - Adjacency matrix for graph storage
 - Priority queue (min-heap) for Dijkstra's efficiency
- **Classes:**
 - graph: Manages routes and operations
 - mst: Handles Prim's algorithm
 - sssp: Handles Dijkstra's algorithm

For the input handling and main.cpp file I used AI to help me with it. Anything else I write it myself.

My plan for the project, and future work to do

